

Markov Chains: What Comes Next?

Lesson Guide | Unplugged Activities

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Activity 1.

What Comes Next?: Introducing Markov Chains

1.1 Objective:

This activity introduces learners to Markov Chains as a machine learning method to predict the next item or token. Using simple sequences of emotion emojis, learners predict the following emotion emoji.

1.2 Introduction:

- Start the activity by reminding learners that artificial intelligence/machine learning uses existing datasets to predict new data.
 - You can build on previous activities you may have done with your learners, including classification tasks that predict if an element belongs to a specific category, or ask learners to think about how AI/ML systems in their everyday life make predictions.
- Introduce Markov Chains as an ML method used to predict what comes next based on what has happened before. Markov Chains can be used to predict events, letters, words, numbers, or emoji that are part of a sequence. We call these elements (events, letters, words, numbers, or emoji) tokens. A **token** is a fundamental unit or element.
- To give some context for why Markov Chains are called Markov Chains, explain that these were first investigated by Andrey Markov, a mathematician and chess player. Markov used chains to model the alliteration of vowels and consonants in Russian literature.

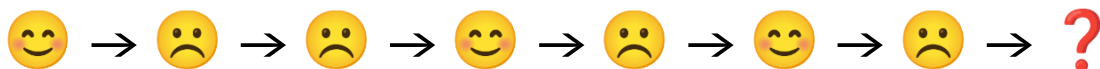
1.3 Doing it together:

- Invite learners to guess what the next token (emoji) will be in the following sequence of emoji:



- This should be pretty intuitive, and most learners will guess that the next token will be a smile face.

- Invite learners to guess what comes next in the following sequence:

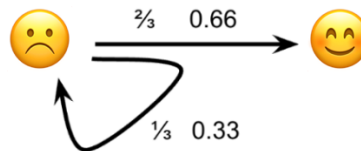


- This time, some learners will guess the next emoji is a sad face and others will guess a happy face. Ask learners to explain their decisions.
- Explain that we can calculate the probability for each possible next emoji and then choose the most probable answer. Show the following diagram:



- Ask:

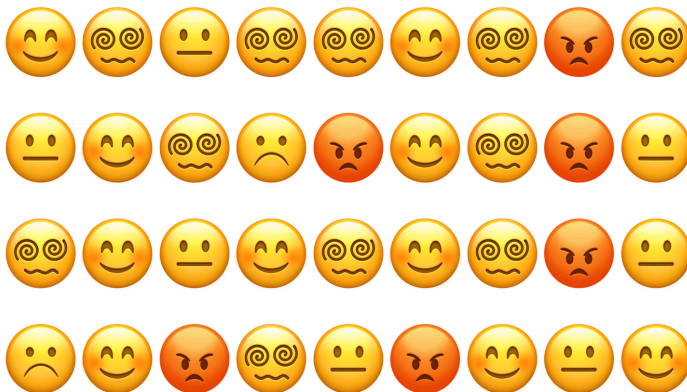
- How many sad faces are followed by a happy face or another sad face? (Answer: 3)
- How many sad faces are followed by a happy face? (Answer: 2)
- How many sad faces are followed by another sad face? (Answer: 1)
- Compute the probability for each possible next token:



- Discuss that it is **most probable** that the next emoji will be a happy face, but it could also be a sad face.
- Introduce the following definition: A Markov chain is a sequence of events where the probability of the future only depends on the present state. Markov chains are simple language models.
 - Explain that the present in the last sequence is a sad face.
 - Explain that a language model computes the probability of a sequence of tokens to calculate the probability of an upcoming token.
- Discuss how Markov Chains may be used in weather predictions, games (and gambling), stock market predictions, predicting behaviors, and analyzing biological and chemical sequences.

1.4 Guessing the next face:

- Individually or in pairs, ask learners to calculate the probabilities for a token following this sequence of emoji:



Answer Key		
😊 → 🌀	6/9	= 0.67
😊 → 😡	1/9	= 0.11
😊 → 😐	2/9	= 0.22
😊 → 😊	0/9	= 0
😊 → 😐	0/9	= 0

Activity 2.

N-imals: Introducing N-grams

2.1 Objective:

This optional activity introduces the concept of n-grams (e.g., bigrams and trigrams) as a way of modifying Markov Chains. Using simple sequences of animal emojis, learners will predict the following animal emoji following the last *n-number* of tokens (last two for bigrams, three for trigrams, etc.).

2.2 Introduction:

- Reiterate Markov Chains using the following sequence:



-  →  = 5/7
-  →  = 2/7
- Introduce the idea of n-grams, or that Markov Chains could also consider *multiple* tokens as the “present state.”
 - A Markov Chain with an n-gram of two (n=2), or a bigram, would predict based on the last *two* tokens. For an n-gram of three (n=3), or a trigram, the prediction is based on the last *three* tokens.
 - Note that the prediction (or *output*) will always be a single token.

2.3 Doing it together:

- For the same sequence, prompt learners to predict what follows the last *two* emojis:



- There are four instances of [ ] before the final two, highlighted below:



- In this case, there is a 0.5 probability of a lion and a 0.5 probability of an elephant after  .

2.4 Guess the N-imal:

- Individually or in pairs, ask learners to calculate the probabilities for a token following this sequence of emoji using a bigram Markov Chain:



Answer Key		
  → 	1/7	= 0.14
  → 	0/7	= 0
  → 	4/7	= 0.57
  → 	0/7	= 0
  → 	2/7	= 0.29

Questions to Ask:

- What are some instances where you would want to use n-gram Markov Chains, as opposed to simple, unigram Markov Chains? (hint: think about words!)

Activity 3.

Sunny, Rainy, or Cloudy: Introducing State Diagrams

3.1 Objective:

This activity introduces state transition diagrams: a visual representation of a system that depicts states as nodes and arrows as transitions, with probabilities indicating how likely the system moves from one state to another. Using a sequence of weather data, learners will create a state diagram and then use it to predict future weather.

3.2 Introduction:

- Introduce transition diagrams as diagrams that visualize the transitions between different states or tokens.
- Note that in previous activities, we observed the probabilities of a single set of tokens (the last n tokens). Transition diagrams allow us to illustrate the probability of **all** possible state transitions of a given collection of weather.

3.3 Doing it together:

- Use the following sequence of emoji to demonstrate how to calculate the probabilities of all state transitions:



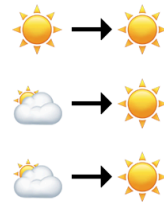
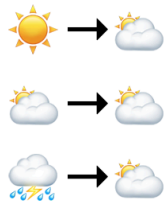
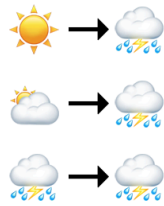
- Prompt learners to list all of the possible transitions in the sequence.
 - Sunny → Sunny; Sunny → Rainy; Rainy → Sunny; Rainy → Rainy
- Calculate the probabilities of each possible transition:
 -  →  = 5/10
 -  →  = 5/10
 -  →  = 4/7
 -  →  = 3/7

3.4 Will it be sunny? rainy? cloudy?

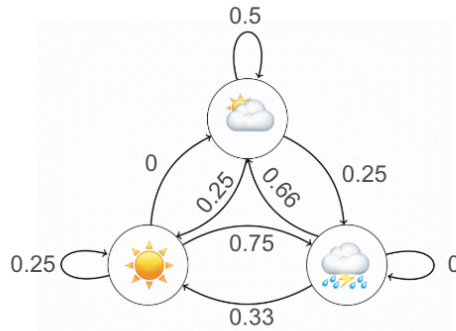
- Use the following sequence of emoji to demonstrate how, with more possible states, transition diagrams become more complex:



- Prompt learners to list all of the possible transitions in the sequence.

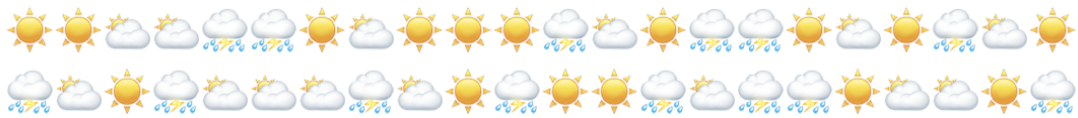


- Calculate the probability of each possible transition in the sequence and complete the transition state diagram:



3.5 On your own:

- Individually or in pairs, ask learners to fill out a three-way transition diagram using the following sequence:



Answer (Fractions)	Answer (Decimals)

Activity 4.

Rock, Paper, Scissors: Rapid Prototyping of Datasets

4.1 Objective:

This activity engages learners in thinking about building datasets and observing how datasets affect the output of a Markov Chain. Using the familiar game of Rock, Paper, Scissors, learners will actively build their own datasets, resulting in a final round determined by the output of their dataset. This activity engages learners in the development of datasets, learning about the impact of having *more* data, and careful considerations for curating data.

4.2 Introduction:

- In the previous activities, the data was provided for you, in the form of sequenced emoji.
- If we were designing a Markov Chain based on the animals above, someone had to provide a sequence of animals in the first place!
- In the real world, data needs to be collected, curated, and labeled so the computer is able to train a machine learning model.
- Here, learners will build their own datasets through a casual game of rock paper scissors!

4.3 Doing it together:

- Note: This game of rock-paper-scissors will be completed in 3 sets of 10-15 rounds (total 30-45 rounds of RPS). The winner of this game is determined by the winner of the final set.
- Optionally, use this webtool: <http://daninoh.github.io/rps-markov>
- Learners will group up in pairs.
- With a piece of paper and a writing utensil in hand, they will play rock-paper-scissors (RPS) against each other 10-15 times. For each round of RPS, learners will note what shape each person played (rock, paper, or scissors).
- By the end of the 10-15 rounds, each learner should have their own sequence of rocks, papers, and scissors. They have created their first dataset!

○ Example:

■ Player 1:                 [?]

■ Player 2:                 [?]

- Using this dataset, learners can predict what would come next.
- Considering this knowledge, learners will iterate on their dataset, by adding (appending) more data through 10-15 more rounds of RPS.

○ Example:

■ [Player 1]

-                 ...
-                

■ [Player 2]

-                  ...
 -                 
- Using this lengthened dataset, learners will predict what shape would come next.
 - The learners will do one final set of rock-paper-scissors, determine the most likely token to come next, compare shapes, and declare a winner!

4.4 Questions to Ask:

- (After the first set) Do you notice any patterns in how you play rock-paper-scissors?
- (Between each set) What will you change about your rock-paper-scissors pattern to try to win?